NOURISH-SSc: A Nutrition and Gastrointestinal Symptom Tracking and Follow-up Prototype for Systemic Sclerosis

NOURISH: Autoimmunity Digital Health Challenge — Phase 1

Participant or Team: [Umar Tabbsum]

1 Problem, Scientific Rationale, and Scope

Systemic sclerosis (SSc), also called scleroderma, is a multisystem autoimmune disease that can affect the skin and internal organs, including the gastrointestinal (GI) tract. Reflux, abdominal distension, swallowing or eating difficulties, and other GI symptoms can interfere with food intake, daily activities, and quality of life. Symptoms and their practical consequences may fluctuate between appointments, making it difficult to recall and communicate patterns consistently.

Published research supports the importance of understanding nutrition-related needs in SSc. Samm et al. described nutrition concerns and perspectives among people living with SSc, including interest in nutrition support [1]. A separate cross-sectional study reported associations between nutritional-status indices and GI symptom burden in people with SSc [2]. These findings support the rationale for documenting symptoms and nutrition-related experiences together. They do not establish that a particular dietary change causes a symptom or that digital tracking improves clinical outcomes.

Proposed solution. NOURISH-SSc is an early-stage web prototype that helps users record GI symptoms and eating-related experiences, review entries over time, track concerns requiring follow-up, and prepare a user-reviewed summary for a potential care conversation. Its workflow connects three practical needs: documenting experiences, recognizing patterns in one's own records, and organizing questions or concerns for follow-up.

The current prototype includes symptom check-ins, saved dated entries, descriptive symptom averages and trends, nutrition-related fields, optional notes, follow-up tracking, and a Care Summary. Users can record concerns such as reflux, bloating or distension, reduced appetite, and eating or swallowing difficulty. Entries can include reported nutrition changes, meal impact, and free-text notes. The follow-up workflow records a concern, its reported impact, a question or next step, and a status. The Care Summary organizes available entries and relevant follow-up items within a selected summary period. Copying, printing, and data export are available as part of the prototype workflow.

These functions organize self-reported information; they do not determine the cause of symptoms or recommend a diet. Users remain responsible for reviewing a summary before deciding whether to share it. The prototype is not a diagnostic, treatment, or clinical decision-support system.

Scientific rationale and measures. NOURISH-SSc focuses on experiences that may be relevant to GI functioning, eating, and nutrition discussions. Its symptom ratings are descriptive self-reports, not validated clinical scores. Averages and timelines summarize what a user has entered; they do not establish disease activity, nutritional status, malnutrition, or clinical risk. The app does not directly measure immune biomarkers, laboratory results, or autoimmune disease activity.

The University of California, Los Angeles Scleroderma Clinical Trial Consortium Gastrointestinal Tract 2.0 (UCLA SCTC GIT 2.0) is an established patient-reported measure of GI symptoms and their impact in SSc [3]. NOURISH-SSc does not reproduce or calculate that instrument and is not a substitute for it. Future research could assess whether the prototype's longitudinal records complement validated questionnaires and, where appropriately collected, clinical measures.

No predictive model has been trained on patient-level data for this prototype. Any literature-derived values used during development must be checked against their original publications; they are not results from NOURISH-SSc users. The prototype's fictional test entries are for demonstrating functionality only.

2 Engagement and Intended Users

The prototype has been developed as an independent, early-stage project informed by published research and publicly available information about SSc. Its design prioritizes a short check-in, a readable history, explicit follow-up status, and a summary that users can inspect before sharing.

Formal patient interviews, clinician consultations, focus groups, co-design sessions, and structured usability studies have not been conducted. The project therefore does not claim patient co-design, clinical endorsement, or validation by people living with SSc or healthcare professionals. Testing to date has focused on the prototype workflow using fictional or synthetic entries, not real patient records.

The intended users are adults living with SSc who experience GI or nutrition-related difficulties and want to organize their own observations. Relevant future stakeholders could include people living with SSc, rheumatology and gastroenterology professionals, registered dietitians, accessibility specialists, and researchers. Their input could help assess whether the fields reflect user priorities, whether the check-in burden is acceptable, and whether the summary is clear without implying medical conclusions.

Any future engagement or human-participant study would require an appropriate plan for consent, privacy, data handling, and oversight. Feedback and resulting design decisions should be documented before claims are made about usability or user acceptance. Until such work is undertaken, these matters remain evaluation priorities, not demonstrated project outcomes.

3 Potential Impact

A dated record of symptoms, eating difficulties, nutrition changes, and follow-up questions could help users prepare for appointments and communicate concerns more consistently. A consolidated summary may make it easier to review experiences across several entries rather than relying entirely on recall during a short consultation.

The immediate potential benefit is better organization of self-reported information. It is not a demonstrated improvement in symptoms, nutritional status, quality of life, referral rates, or other health outcomes. The prototype does not decide whether a referral is needed; users and qualified professionals retain responsibility for clinical decisions.

The current implementation demonstrates a workflow from check-in to timeline, follow-up tracking, and a user-prepared visit summary. Fictional test data can demonstrate how information is displayed and carried into a summary, but cannot establish clinical usefulness or sustained use. No formal beta test, clinical pilot, outcomes study, or patient-derived usability assessment has been completed.

If the project advances to evaluation, initial measures could include task completion, time required to complete a check-in, missing or confusing fields, perceived burden, comprehension of the summary, and perceived usefulness. Further feasibility work could examine whether users can find and update follow-up items and correctly understand that symptom averages are descriptive rather than clinical thresholds. Any evaluation involving participants would require appropriate safeguards and oversight.

Sustained use would depend on keeping entry brief, making past records easy to review, ensuring accessible presentation, and giving users control over data sharing and deletion. The prototype's current local-storage

approach is suitable only for demonstration with fictional or synthetic information, not for storing real sensitive health records.

4 Feasibility and Implementation

NOURISH-SSc is a functional web-based prototype demonstrating the core user workflow. Its implemented functions include recording and reviewing symptom entries, displaying descriptive trends, documenting nutrition-related experiences, managing follow-up concerns, and preparing a Care Summary that can include relevant follow-up items. The prototype has been exercised using fictional test entries. This demonstrates aspects of the user interface and workflow, not formal reliability, security, accessibility, or clinical validity.

The current demonstration stores information in the browser. It is not a secure patient-record system. Local browser storage does not provide the safeguards expected of a production health application, and users should not enter real sensitive health information. The app does not provide diagnosis, treatment recommendations, automated malnutrition scoring, biomarker interpretation, or clinical risk prediction.

Further work can proceed in stages:

1. **Workflow refinement.** Review the clarity and consistency of symptom labels, nutrition-related fields, follow-up statuses, date filtering, summaries, and export behavior. Continue testing with fictional data and document defects and corrections.
2. **Privacy and security architecture.** Before any real health information is collected, define data flows, access controls, authentication where required, appropriate encryption, retention and deletion rules, backup and recovery procedures, and the user's control over sharing. Complete a security review before production use.
3. **Accessibility and usability evaluation.** Assess mobile usability, readable text, contrast, plain-language instructions, keyboard and assistive-technology access, and error recovery. Any formal participant evaluation should use an appropriate protocol and oversight.
4. **Feasibility and clinical-context evaluation.** If justified by early findings, examine whether users can complete the workflow reliably and whether the summary complements, rather than replaces, established patient-reported measures and professional assessment. Define outcomes before conducting an evaluation.
5. **Deployment and maintenance.** Establish responsibility for hosting, updates, security monitoring, accessibility maintenance, support, and long-term operating costs. Test compatibility across supported devices and browsers before broader deployment.

A plausible initial workflow would let users review a summary and choose whether to share it themselves, without requiring electronic health record integration. No such integration currently exists. Future interoperability would require separate technical, privacy, consent, and institutional review.

The initial scope is SSc, where GI and nutrition-related difficulties provide a focused use case. The general workflow might eventually be adapted to other autoimmune conditions, but each adaptation would require condition-specific evidence, stakeholder input, and evaluation of relevant symptoms and measures.

5 Innovation and Differentiation

Symptom diaries, food trackers, timelines, and visit summaries are not new in themselves. NOURISH-SSc focuses on connecting SSc-relevant GI and eating experiences with follow-up tracking and a user-reviewed summary. Its intended contribution is a practical workflow linking daily observations to questions and concerns a user may wish to discuss with a care team.

The prototype combines descriptive symptom history, nutrition-related context, follow-up status, and summary preparation in one workflow. This combination is a design hypothesis, not proof of novelty or superiority. A systematic comparison with existing SSc-specific and general symptom-tracking tools would be needed to establish the extent of differentiation.

The approach is deliberately conservative. It does not infer food triggers, prescribe elimination diets, generate a malnutrition diagnosis, interpret laboratory biomarkers, or make clinical decisions. Instead, it organizes what the user reports and makes the limitations of the information explicit.

If future feasibility work supports the approach, structured longitudinal records could be studied as a way to describe changes in self-reported symptoms and eating experiences. Validated instruments and appropriately collected clinical measures would be needed for relevant comparisons. Observational records alone cannot establish causal relationships between foods, dietary changes, and symptoms.

NOURISH-SSc is therefore best characterized as an early-stage, evidence-informed prototype addressing a focused communication and self-management need. Its next milestone is not to claim clinical efficacy, but to establish usability, clarify the value of its follow-up workflow, and determine whether further evaluation is warranted.

AI Use Disclosure

Artificial intelligence tools assisted with generating voice for presentation. The participant is responsible for reviewing the final submission, checking factual claims and references, and ensuring that the description accurately reflects the work completed. AI-generated content has not been treated as clinical evidence. No AI-generated clinical predictions or patient outcomes are claimed.

References

1. Samm, D.-A., Macoustra, A., Crane, R., McWilliams, L., Proudman, S., and Chapple, L.-A. S. (2024). Views of nutrition needs in patients with systemic sclerosis. *Journal of Scleroderma and Related Disorders, 9*(3), 216–222. https://doi.org/10.1177/23971983241264868

2. Öz, N., Gezer, H. H., Karabulut, Y., and Duruöz, M. T. (2025). Association of nutritional status indices with gastrointestinal symptoms in systemic sclerosis: A cross-sectional study. *Rheumatology International, 45*(1), article 21. https://doi.org/10.1007/s00296-024-05783-2

3. Khanna, D., Hays, R. D., Maranian, P., Seibold, J. R., Impens, A., Mayes, M. D., et al. (2009). Reliability and validity of the University of California, Los Angeles Scleroderma Clinical Trial Consortium Gastrointestinal Tract Instrument. *Arthritis & Rheumatism, 61*(9), 1257–1263. https://doi.org/10.1002/art.24730